Midas

Monocular Depth Estimation

Towards Robust Monocular Depth Estimation: Mixing Datasets for Zero-shot Cross-dataset Transfer

René Ranftl*, Katrin Lasinger*, David Hafner, Konrad Schindler, and Vladlen Koltun

Abstract—The success of monocular depth estimation relies on large and diverse training sets. Due to the challenges associated with acquiring dense ground-truth depth across different environments at scale, a number of datasets with distinct characteristics and biases have emerged. We develop tools that enable mixing multiple datasets during training, even if their annotations are incompatible. In particular, we propose a robust training objective that is invariant to changes in depth range and scale, advocate the use of principled multi-objective learning to combine data from different sources, and highlight the importance of pretraining encoders on auxiliary tasks. Armed with these tools, we experiment with five diverse training datasets, including a new, massive data source: 3D films. To demonstrate the generalization power of our approach we use *zero-shot cross-dataset transfer*, i.e. we evaluate on datasets that were not seen during training. The experiments confirm that mixing data from complementary sources greatly improves monocular depth estimation. Our approach clearly outperforms competing methods across diverse datasets, setting a new state of the art for monocular depth estimation.

Index Terms—Monocular depth estimation, Single-image depth prediction, Zero-shot cross-dataset transfer, Multi-dataset training

MiDaS v3.1 – A Model Zoo for Robust Monocular Relative Depth Estimation

Reiner Birkl, Diana Wofk, Matthias Müller

Intel Labs

2023

Abstract

We release MiDaS v3.1[1] *for monocular depth estimation, offering a variety of new models based on different encoder*

Many deep learning models for depth estimation adopt encoder-decoder architectures. In addition to convolutional encoders used in the past, a new category of encoder options has emerged with transformers for computer vision.

Published in Image Processing On Line on 2023–01–27.
Submitted on 2023–01–10, accepted on 2023–01–10.
ISSN 2105–1232
This article is available online with supplementary materials, software, datasets and online demo at
https://doi.org/10.5201/ipol.2023.459

Monocular Depth Estimation: a Review of the 2022 State of the Art

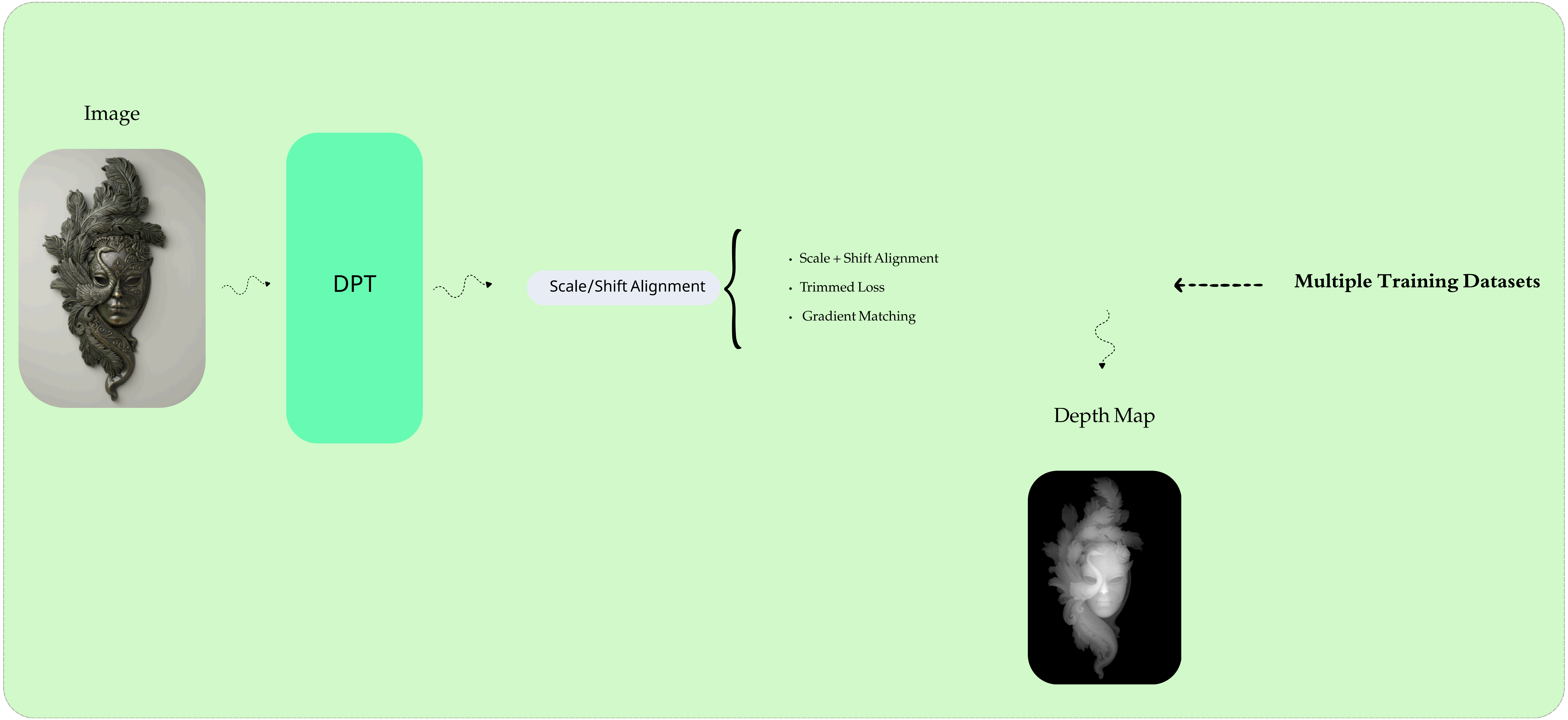
Thibaud Ehret

Université Paris-Saclay, ENS Paris-Saclay, Centre Borelli, Gif-sur-Yvette, France
thibaud.ehret@ens-paris-saclay.fr

Communicated by Jean-Michel Morel *Demo edited by* Thibaud Ehret


v21_small_256
v21_384
dpt_beit_large_512
dpt_beit_large_384
dpt_beit_base_384
dpt_swin2_large_384
dpt_swin2_base_384
dpt_swin2_tiny_256
dpt_swin_large_384
dpt_levit_224
dpt_large_384

MiDas Architecture



Vision Transformers for Dense Prediction

René Ranftl   Alexey Bochkovskiy   Vladlen Koltun

Intel Labs

rene.ranftl@intel.com

Abstract

We introduce dense vision transformers, an architecture that leverages vision transformers in place of convolutional

on the decoder and its aggregation strategy [6, 7, 50, 53]. However, it is widely recognized that the choice of backbone architecture has a large influence on the capabilities of the overall model, as any information that is lost in the

تبدیل یک Vision Transformer مخصوص classification

Vision Transformers for Dense Prediction

René Ranftl   Alexey Bochkovskiy   Vladlen Koltun

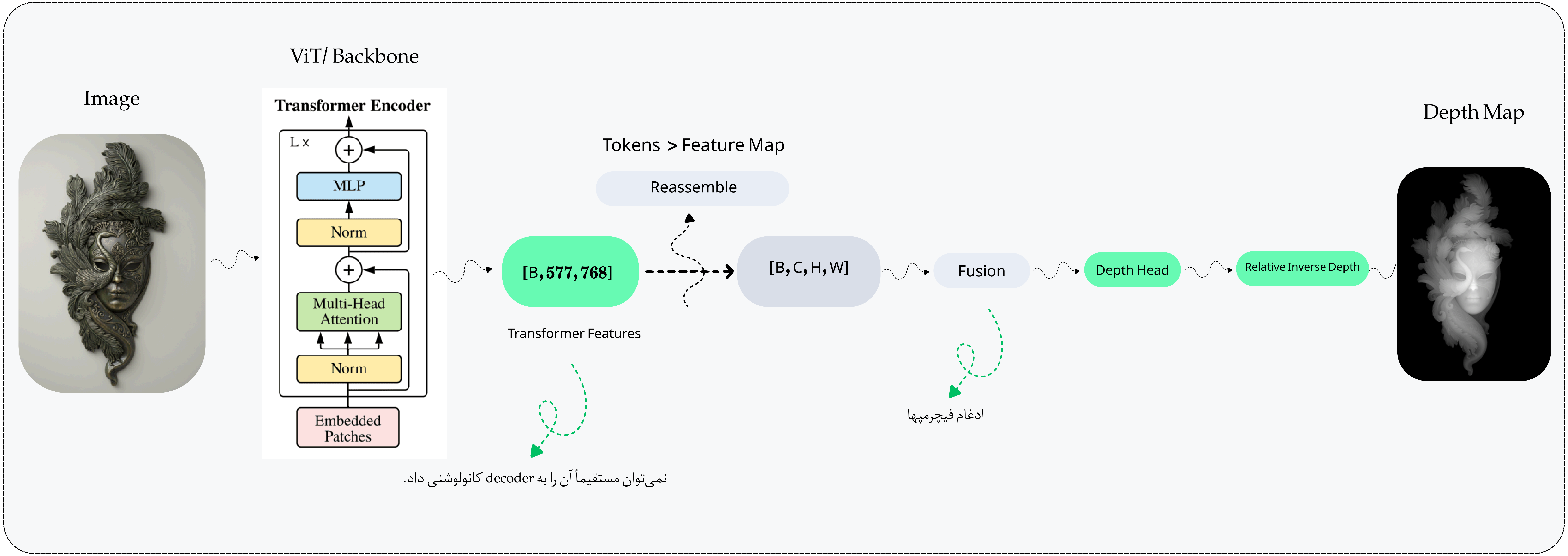
Intel Labs

rene.ranftl@intel.com

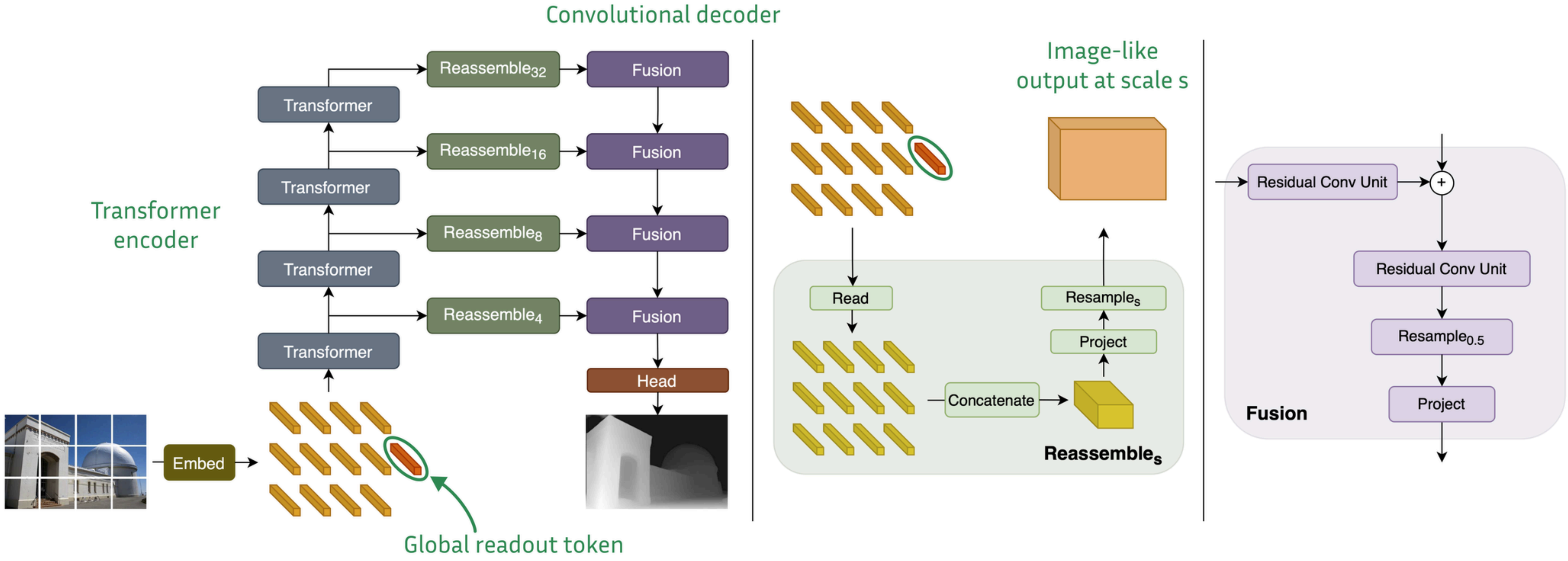
Vision Transformer مخصوص dense prediction

DPT Architecture

Vision Transformers for Dense Prediction (DPT)



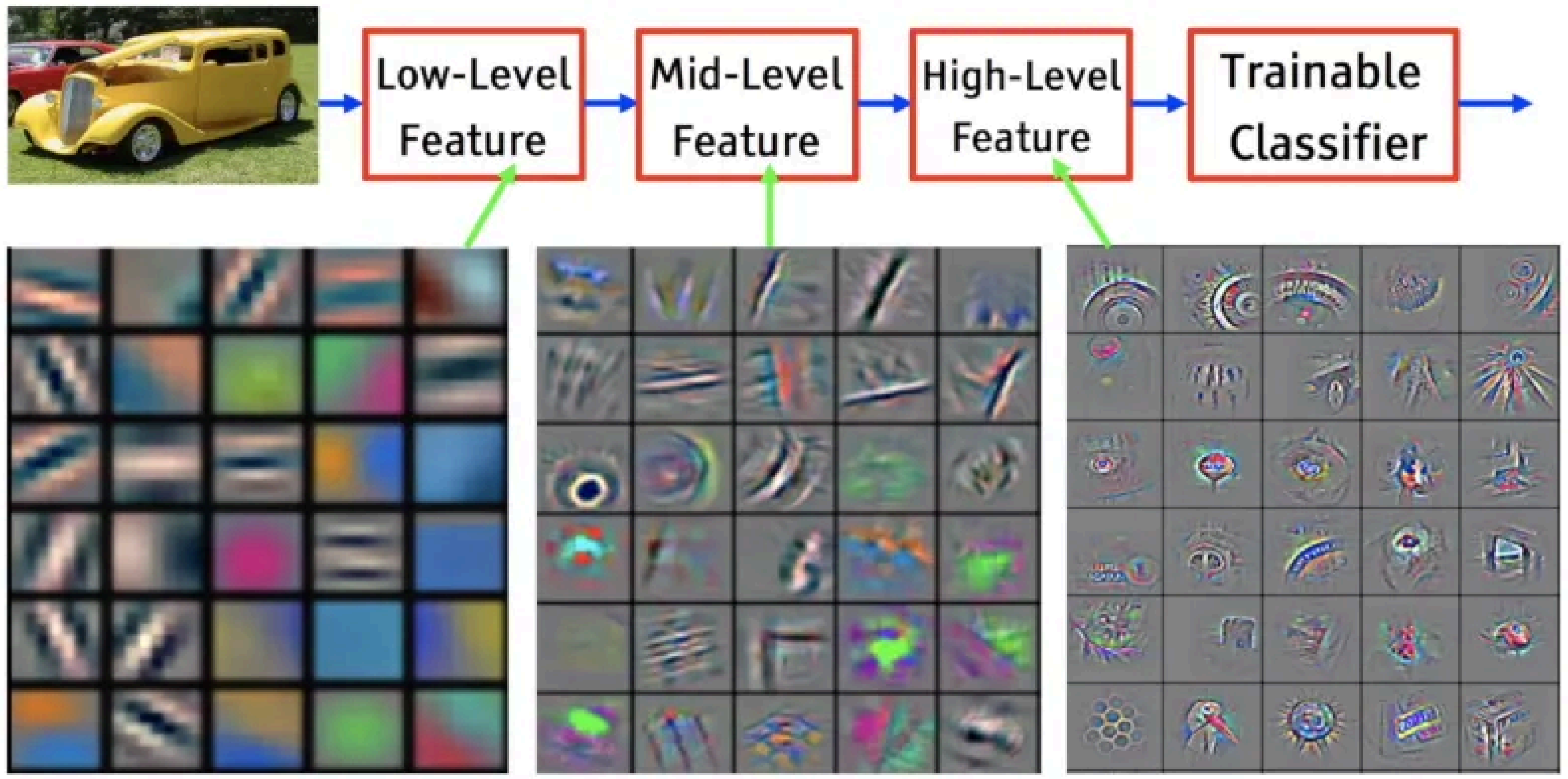
Vision Transformers for Dense Prediction (DPT)

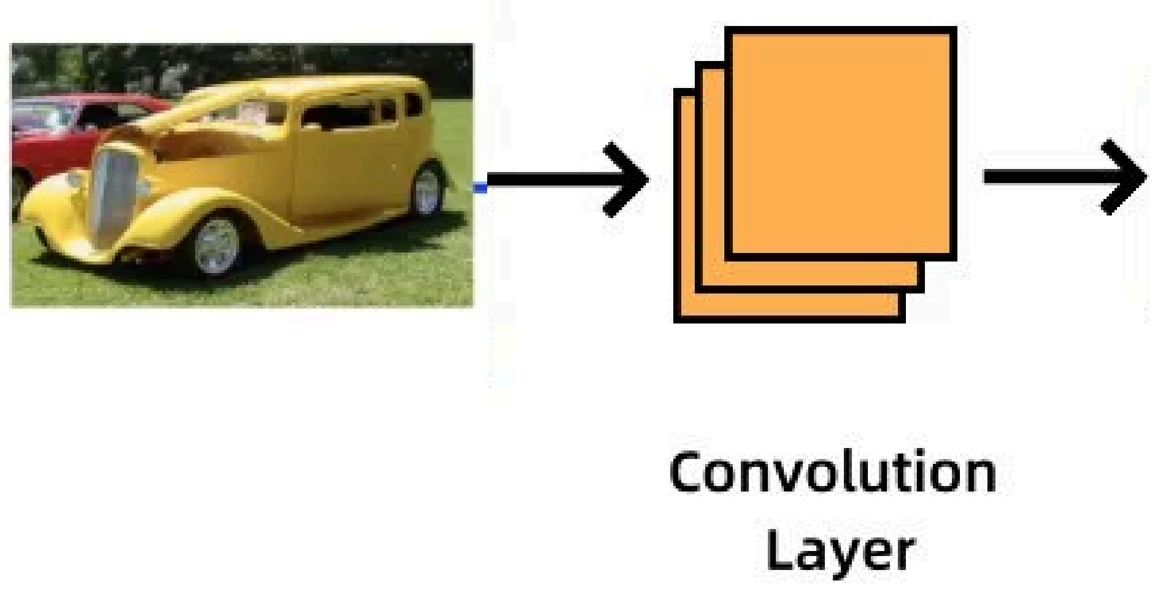


خروجی ViT اصلاً Feature Map ندارد

Reassemble

خروجی CNN





Feature maps

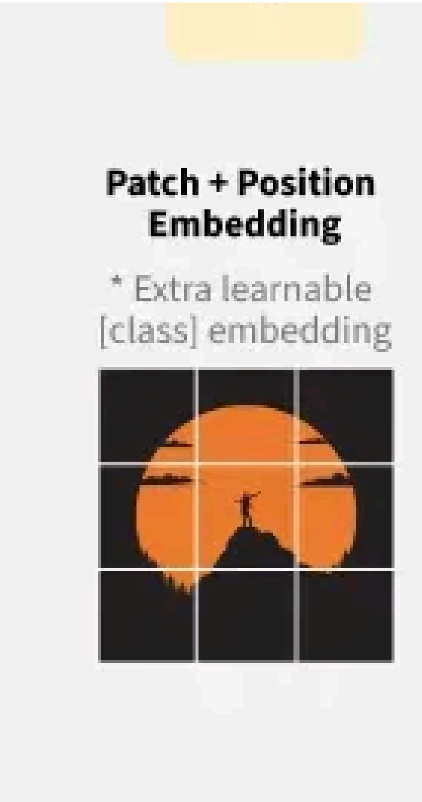
(B, C, H, W)

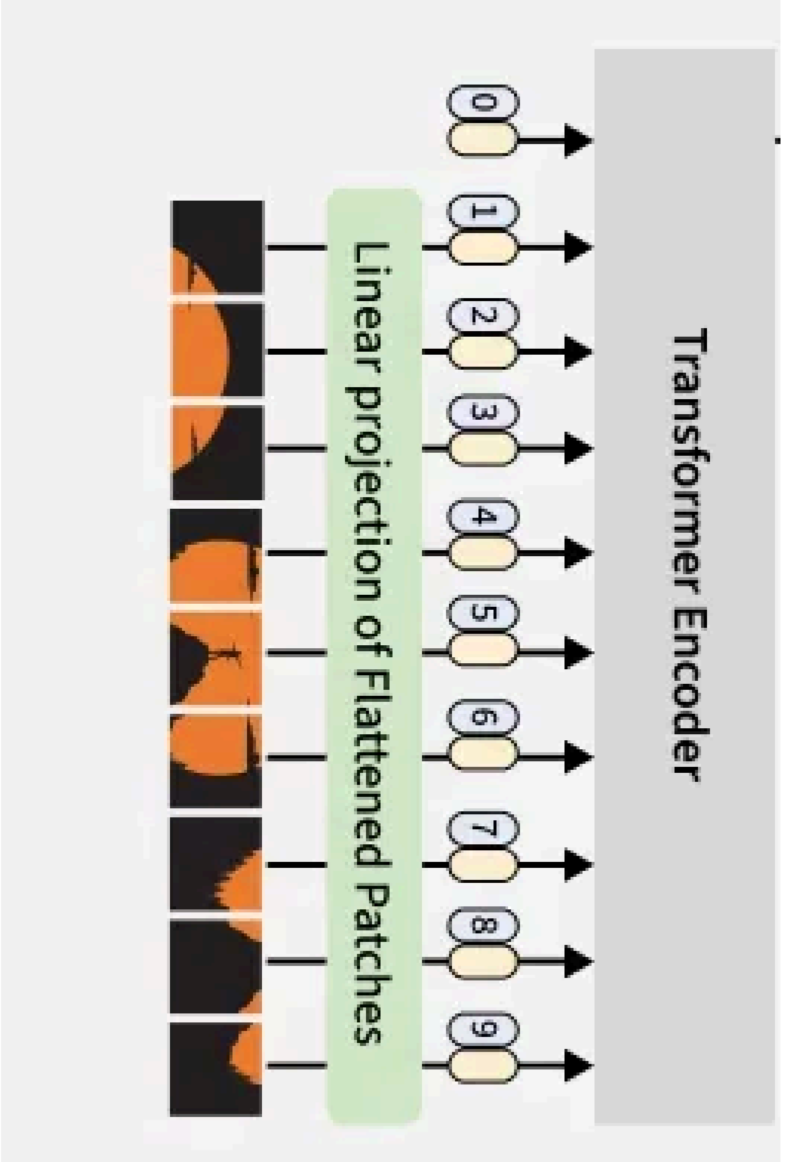
اطلاعات مکانی هنوز به شکل یک grid دوبعدی

بسیار مناسب برای segmentation، depth estimation و detection

Reassemble

خروجی VIT



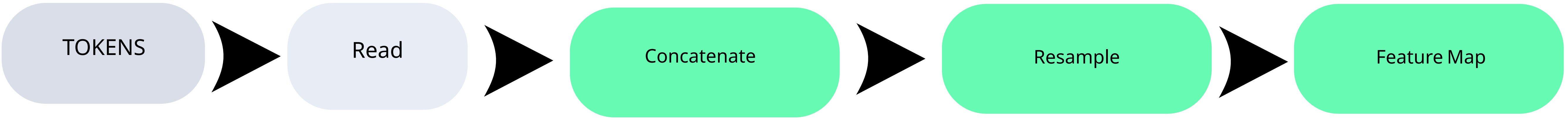


TOKENS
[B, 576, 768]

576 tokens

هر token یک بردار 768 بعدی

Reassemble



Reassemble

[B, 577, 768]

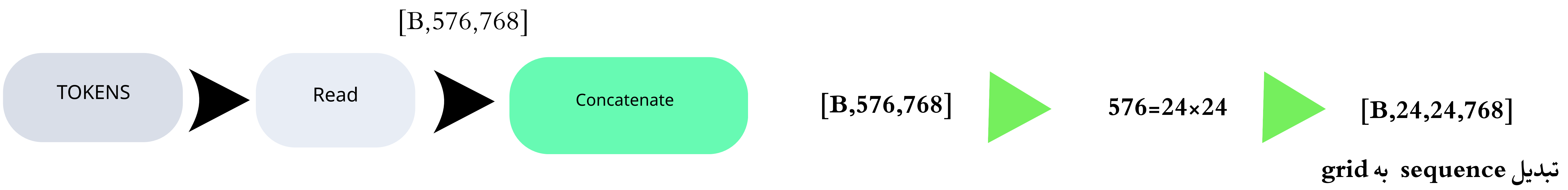
TOKENS → Read

576 patch tokens + 1 readout token = 577

[B,577,768]
↓
[B,576,768]

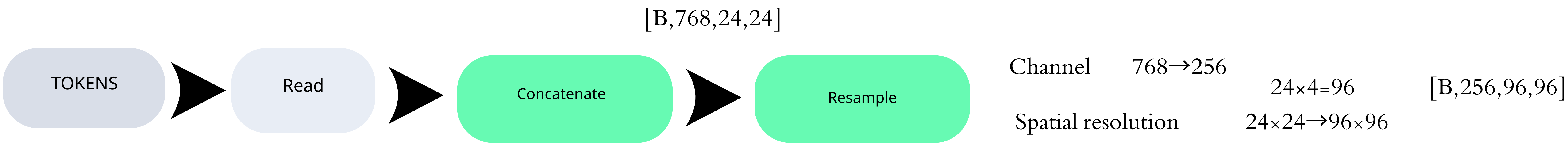
[B,576,768]

Reassemble



TOKENS
Read
[B,576,768]
Concatenate
[B,576,768]
576=24×24
[B,24,24,768]
تبدیل sequence به grid

Reassemble


TOKENS
Read
Concatenate
[B,768,24,24]
Resample
Channel 768→256
Spatial resolution 24×24→96×96
24×4=96
[B,256,96,96]

Fusion

لایه های خروجی vit	Read	Concatenate/reshape	Resample
L3 [B,577,768]	L3 → [B,576,768]	L3 → [B,768,24,24]	L3 → [B,256,96,96]
L6 [B,577,768]	L6 → [B,576,768]	L6 → [B,768,24,24]	L6 → [B,256,48,48]
L9 [B,577,768]	L9 → [B,576,768]	L9 → [B,768,24,24]	L9 → [B,256,24,24]
L12 [B,577,768]	L12 → [B,576,768]	L12 → [B,768,24,24]	L12 → [B,256,12,12]

Fusion

reassmble خروجی

F1 : [B, 256, 96, 96]
F2 : [B, 256, 48, 48]
F3 : [B, 256, 24, 24]
F4 : [B, 256, 12, 12]

$$((F4 *2 + F3)*2 + F2)*2 + F1$$

Depth Head

$$\mathbf{F \rightarrow d(x,y)}$$

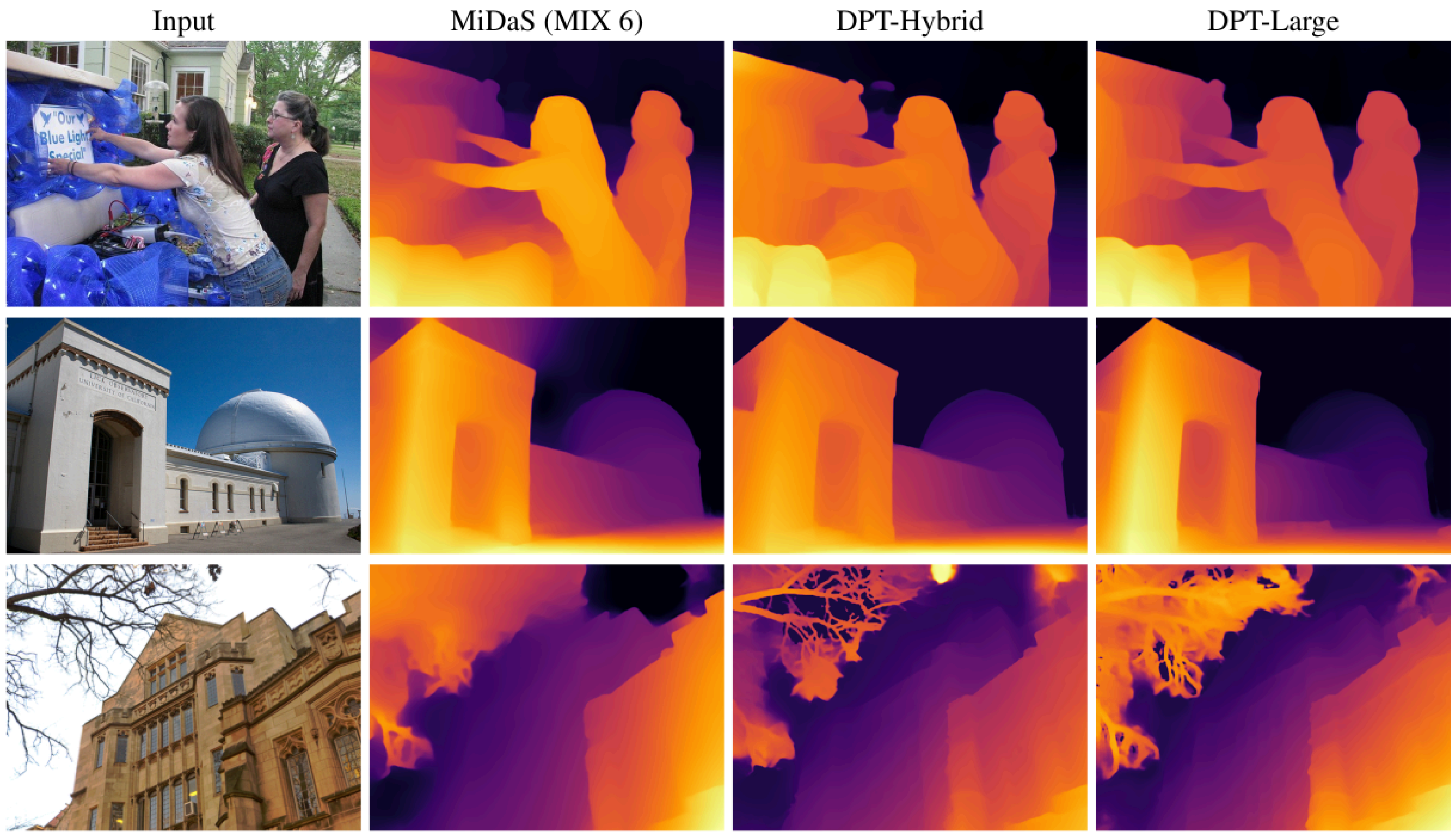


Figure 2. Sample results for monocular depth estimation. Compared to the fully-convolutional network used by MiDaS, DPT shows better global coherence (e.g., sky, second row) and finer-grained details (e.g., tree branches, last row).

	Training set	DIW WHDR		ETH3D AbsRel		Sintel AbsRel		KITTI δ>1.25		NYU δ>1.25		TUM δ>1.25	
DPT - Large	MIX 6	**10.82**	(-13.2%)	**0.089**	(-31.2%)	**0.270**	(-17.5%)	**8.46**	(-64.6%)	**8.32**	(-12.9%)	**9.97**	(-30.3%)
DPT - Hybrid	MIX 6	11.06	(-11.2%)	0.093	(-27.6%)	0.274	(-16.2%)	11.56	(-51.6%)	8.69	(-9.0%)	10.89	(-23.2%)
MiDaS	MIX 6	12.95	(+3.9%)	0.116	(-10.5%)	0.329	(+0.5%)	16.08	(-32.7%)	8.71	(-8.8%)	12.51	(-12.5%)
MiDaS [30]	MIX 5	12.46		0.129		0.327		23.90		9.55		14.29	
Li [22]	MD [22]	23.15		0.181		0.385		36.29		27.52		29.54	
Li [21]	MC [21]	26.52		0.183		0.405		47.94		18.57		17.71	
Wang [40]	WS [40]	19.09		0.205		0.390		31.92		29.57		20.18	
Xian [45]	RW [45]	14.59		0.186		0.422		34.08		27.00		25.02	
Casser [5]	CS [8]	32.80		0.235		0.422		21.15		39.58		37.18	

Table 1. Comparison to the state of the art on monocular depth estimation. We evaluate zero-shot cross-dataset transfer according to the protocol defined in [30]. Relative performance is computed with respect to the original MiDaS model [30]. Lower is better for all metrics.

	δ>1.25	δ>1.25^2	δ>1.25^3	AbsRel	RMSE	log10
DORN [13]	0.828	0.965	0.992	0.115	0.509	0.051
VNL [48]	0.875	0.976	0.994	0.111	0.416	0.048
BTS [20]	0.885	0.978	0.994	**0.110**	0.392	0.047
DPT-Hybrid	**0.904**	**0.988**	**0.998**	**0.110**	**0.357**	**0.045**

Table 2. Evaluation on NYUv2 depth.

	δ>1.25	δ>1.25^2	δ>1.25^3	AbsRel	RMSE	RMSE log
DORN [13]	0.932	0.984	0.994	0.072	2.626	0.120
VNL [48]	0.938	0.990	0.998	0.072	3.258	0.117
BTS [20]	0.956	0.993	0.998	**0.059**	2.756	0.096
DPT-Hybrid	**0.959**	**0.995**	**0.999**	0.062	**2.573**	**0.092**

Table 3. Evaluation on KITTI (Eigen split).

MiDaS v3.1 – A Model Zoo for Robust Monocular Relative Depth Estimation

Reiner Birkl, Diana Wofk, Matthias Müller

Intel Labs

Abstract

We release MiDaS v3.1[1] for monocular depth estimation, offering a variety of new models based on different encoder

Many deep learning models for depth estimation adopt encoder-decoder architectures. In addition to convolutional encoders used in the past, a new category of encoder options has emerged with transformers for computer vision.

Model Encoder/Backbone	Data Mix	Resources Par. ↓	 FPS ↑	Unconstrained Resolution DIW WHDR ↓	 ETH3D REL ↓	 Sintel REL ↓	 KITTI δ_1 ↓	 NYU δ_1 ↓	 TUM δ_1 ↓	 I %↑	Square Resolution DIW WHDR ↓	 ETH3D REL ↓	 Sintel REL ↓	 KITTI δ_1 ↓	 NYU δ_1 ↓	 TUM δ_1 ↓	 I %↑
BEiT$_{512}$-L [26]	5+12	345	5.7	0.114	**0.066**	0.237	11.57*	**1.862***	**6.132**	19	0.112	**0.061**	**0.209**	**5.005***	**1.902***	**6.465**	**36**
BEiT$_{384}$-L [26]	5+12	344	13	0.124	0.067	0.255	9.847*	2.212*	7.176	16.8	0.111	0.064	0.222	5.110*	2.229*	7.453	33.0
BEiT$_{512}$-L@384 [26]	5+12	345	5.7	0.125	0.068	**0.218**	**6.283***	2.161*	**6.132**	**28**	0.117	0.070	0.223	6.545*	2.582*	6.804	29
SwinV2-L [30]	5+12	213	41	–	–	–	–	–	–	–	0.111	0.073	0.244	5.840*	2.929*	8.876	25
SwinV2-B [30]	5+12	102	39	–	–	–	–	–	–	–	0.110	0.079	0.240	5.976*	3.284*	8.933	23
Swin-L [29]	5+12	213	49	–	–	–	–	–	–	–	0.113	0.085	0.243	6.601*	3.343*	8.750	21
BEiT$_{384}$-B [26]	5+12	112	31	0.116	0.097	0.290	26.60*	3.919*	9.884	-31	0.114	0.085	0.250	8.180*	3.588*	9.276	16
Next-ViT-L-1K-6M [32]	5+12	72	30	**0.103**	0.095	0.230	6.895*	3.479*	9.215	16	**0.106**	0.093	0.254	8.842*	3.442*	9.831	14
ViT-L [13]	3+10	344	61	0.108	0.089	0.270	8.461	8.318	9.966	0	0.112	0.091	0.286	9.173	8.557	10.16	0
ViT-B Hybrid [13]	3+10	123	61	0.110	0.093	0.274	11.56	8.69	10.89	-10	-	-	-	-	-	-	-
SwinV2-T [30]	5+12	42	64	–	–	–	–	–	–	–	0.121	0.111	0.287	10.13*	5.553*	13.43	-6
ResNeXt-101 [38]	3+10	105	47	0.130	0.116	0.329	16.08	8.71	12.51	-32	-	-	-	-	-	-	-
LeViT-224 [33]	5+12	51	73	–	–	–	–	–	–	–	0.131	0.121	0.315	15.27*	8.642*	18.21	-34
EfficientNet-Lite3 [36]	3+10	**21**	**90**	0.134	0.134	0.337	29.27	13.43	14.53	-75	-	-	-	-	-	-	-

Table 1. **Evaluation of released models (post second training stage).** The table shows the validation of the second training stage (see Sec. 3.3) for the models released in MiDaS v3.1. The dataset definitions 3+10 and 5+12 used for the training can be found in Sec. 3.3. The resources required per model are given by the number of parameters in million (Par.) and the frames per second (FPS, if possible for the unconstrained resolution). The validation is done on the datasets DIW [54], ETH3D [55], Sintel [56], KITTI [53], NYU Depth v2 [52] and TUM [57] with the validation errors as described in Sec. 4.1. The resolution is either unconstrained, *i.e*. the aspect ratio is given by the images in the dataset, or the images are converted to a square resolution. Overall model quality is given by the relative improvement I with respect to ViT-L (cf. Eq. (1)). Note that Next-ViT-L-1K-6M and ResNeXt-101 are short forms of Next-ViT-L ImageNet-1K-6M and ResNeXt-101 32x8d. The suffix @384 means that the model is validated at the inference resolution 384x384 (differing from the training resolution). Legacy models from MiDaS v3.0 and 2.1 are in italics, where ResNeXt-101=midas_v21_384 and Efficientnet-lite3=midas_v21_256_small. Validation errors that could not be evaluated, because of the model not supporting the respective resolution are marked by –. Quantities not evaluated due to other reasons are given by -. The asterisk * refers to non-zero-shot errors, because of the training on KITTI and NYU Depth v2. The rows are ordered such that models with better relative improvement values for the square resolution are at the top. The best numbers per column are bold and second best underlined.

Model		Resources		Unconstrained Resolution							Square Resolution						
	Data	Par.	FPS	DIW	ETH3D	Sintel	KITTI	NYU	TUM	I	DIW	ETH3D	Sintel	KITTI	NYU	TUM	I
Encoder/Backbone	Mix	↓	↑	WHDR↓	REL↓	REL↓	δ_1 ↓	δ_1 ↓	δ_1 ↓	%↑	WHDR↓	REL↓	REL↓	δ_1 ↓	δ_1 ↓	δ_1 ↓	%↑
Swin-L [30]	3+10	213	41	–	–	–	–	–	–	–	0.115	0.086	0.246	12.15	6.571	9.745	2
Swin-T [30]	3+10	42	71	–	–	–	–	–	–	–	0.131	0.120	0.334	15.66	12.69	14.56	-38
MobileViTv2-0.5 [35]	5+12	**13**	**72**	0.430	0.268	0.418	51.77*	45.32*	39.33	-301	0.509	0.263	0.422	37.67*	48.65*	40.63	-286
MobileViTv2-2.0 [35]	5+12	34	61	0.509	0.263	0.422	37.67*	48.65*	40.63	-294	0.501	0.269	0.433	59.94*	48.32*	41.79	-320
BEiT$_{384}$-L 5K+12K	·K	344	13	0.120	0.066	0.213	2.967*	2.235*	**6.570**	35	0.110	0.066	**0.212**	5.929*	2.296*	**6.772**	**33**
BEiT$_{384}$-L Wide	5+12	344	13	0.111	0.068	0.247	10.73*	2.146*	7.217	17.4	0.112	0.066	0.221	**5.078***	2.216*	7.401	**32.7**
BEiT$_{384}$-L 5+12+12K	+12K	344	13	0.123	0.065	0.216	2.967*	2.066*	7.417	33	**0.107**	**0.064**	0.217	5.631*	2.259*	7.659	32
BEiT$_{384}$-L A5+12A	·A	344	13	**0.110**	**0.061**	**0.207**	**2.802***	**1.891***	7.533	**37**	0.113	0.070	0.213	6.504*	**2.179***	7.946	29

Table 2. **Evaluation of unpublished models (post second training stage).** The table shows the validation of the second training stage (see Sec. 3.3) of models not released in MiDaS v3.1 due to a low depth estimation quality. The models below the horizontal separator are based on experimental modifications explained in Sec. 4.3. The general table layout is similar to Tab. 1. The extra dataset mixes, like ·K, are explained in Sec. 4.3.

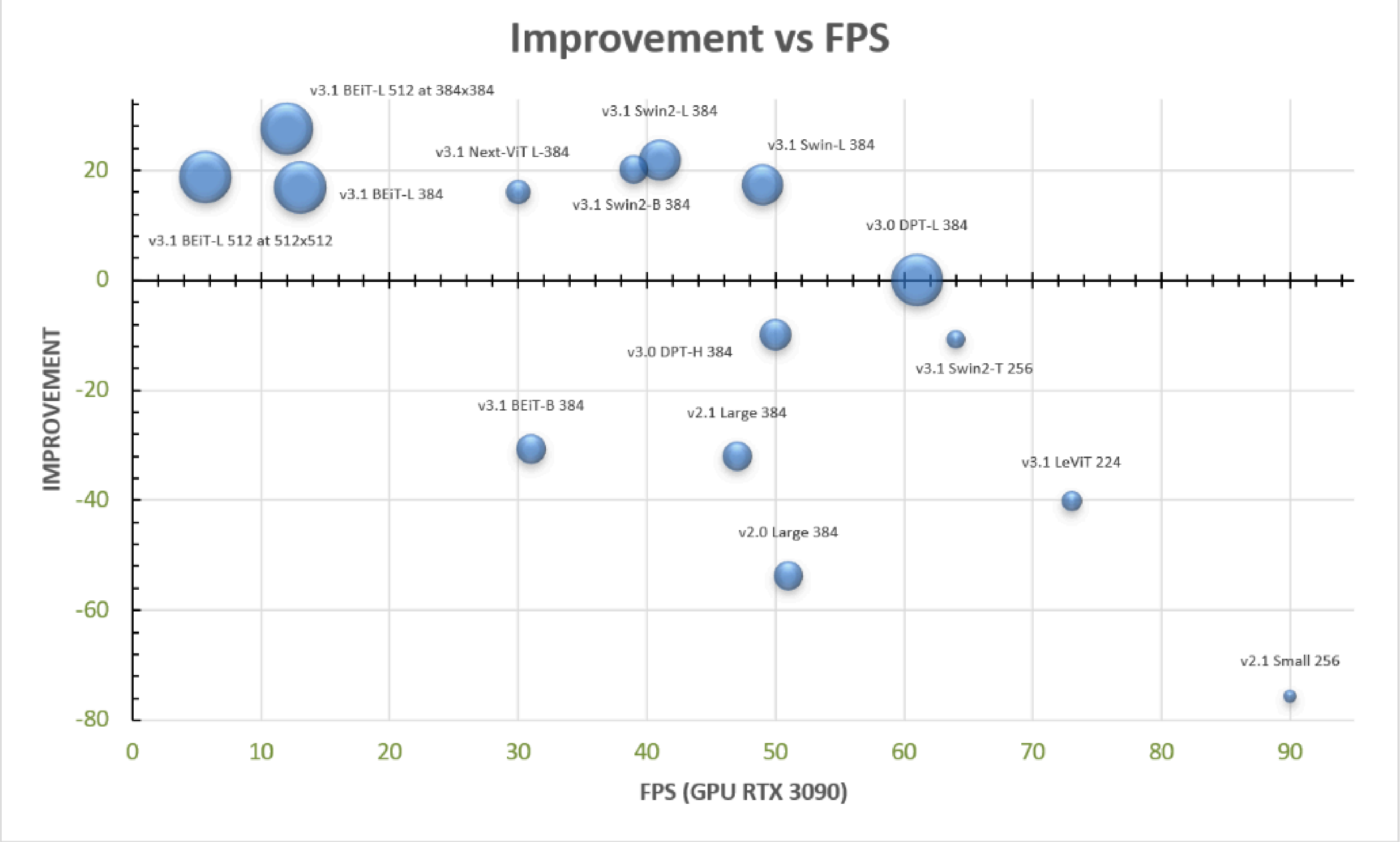


Figure 1. **Improvement vs FPS.** The plot shows the improvement of all the models of MiDaS v3.1 with respect to the largest model $\text{DPT}_{\text{L 384}}$ (=ViT-L 384) of MiDaS v3.0 vs the frames per second. The framerate is measured on an RTX 3090 GPU. The area covered by the bubbles is proportional to the number of parameters of the corresponding models. In the model descriptions, we provide the MiDaS version, because some models of MiDaS v3.1 are legacy models which were already introduced in earlier MiDaS releases. The first 3-digit number in the model name reflects the training resolution which is always a square resolution. For two BEiT models, we also provide the inference resolution at the end of the model description, because there the inference resolution differs from the training one. The improvement is defined as the relative zero-shot error averaged over six datasets as explained in Sec. 4.1.

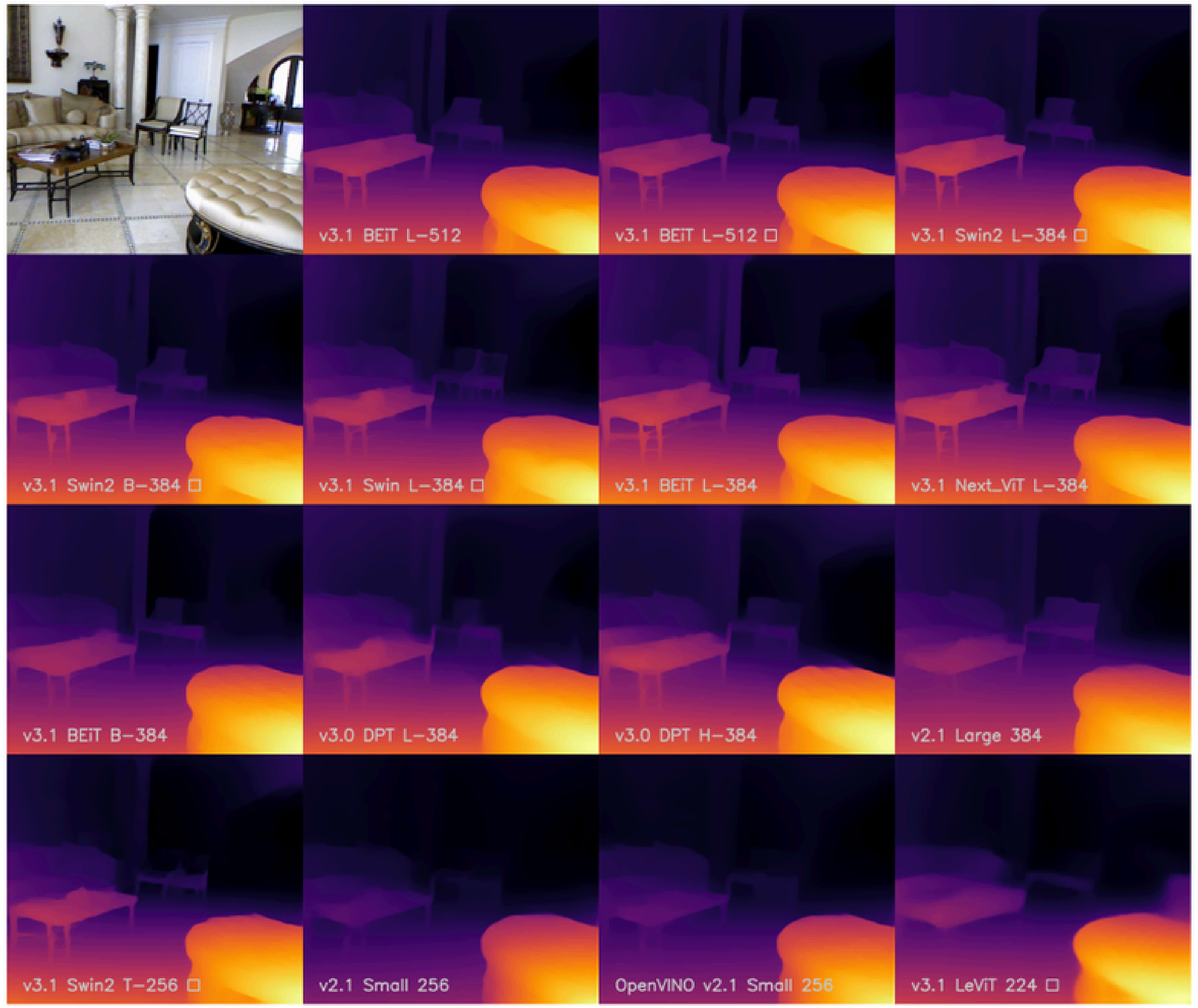


Figure 2. **Backbone comparison.** The table shows the inverse relative depth maps of the different models of MiDaS v3.1, including legacy models, for the example RGB input image at the top, left. The brighter the colors, the larger the inverse relative depths, *i.e.*, the closer the represented objects are to the camera. The names of the models are shown at the bottom left part of each depth map. This includes the MiDaS version, the backbone name and size as well as the training resolution. Models which are evaluated only at a square resolution are marked by the square symbol at the end of the white texts. The second last model at the bottom row is an OpenVINO model.

Model	Square Resolution HRWSI RMSE↓	BlendedMVS REL↓	ReDWeb RMSE↓
BEiT$_{384}$-L [26]	**0.068**	**0.070**	**0.076**
Swin-L [29] Training 1	0.0708	<u>0.0724</u>	0.0826
Training 2	0.0713	<u>0.0720</u>	0.0831
ViT-L [13]	0.071	<u>0.072</u>	0.082
Next-ViT-L-1K-6M [32]	0.075	0.073	0.085
DeiT3-L-22K-1K [34]	<u>0.070</u>	**0.070**	<u>0.080</u>
ViT-L Hybrid [13]	0.075	0.075	0.085
Next-ViT-L-1K [32]	0.078	0.075	0.087
DeiT3-L [34]	0.077	0.075	0.087
ConvNeXt-XL [14]	0.075	0.075	0.085
ConvNeXt-L [14]	0.076	0.076	0.087
EfficientNet-L2 [15]	0.165	0.227	0.219
ViT-L Reversed	0.071	0.073	0.081
Swin-L Equidistant	0.072	0.074	0.083

Table 3. **Model evaluation (post first training stage).** The table shows the validation of unpublished models which were mostly trained only in the first training stage and not also the second one due to low depth estimation quality (see Sec. 3.3). The models above the horizontal separator line (between Next-ViT-L-1K-6M and DeiT3-L-22K-1K) are included for a comparison with the other models and have at least a released variant in Tab. 1, although they were also not released directly (see Sec. 4.2 for details). For Swin-L, two different training runs are shown. The models above the dashed separator are models based on transformer backbones, and the models between the dashed and dotted line are convolutional ones. The rows below the dotted separator are models with experimental modifications as explained in Sec. 4.3. All the models in this table are trained on the 3+10 dataset configuration (in contrast to the mixtures of Tabs. 1 and 2). Validation is done on the datasets HRWSI [48], BlendedMVS [50] and ReDWeb [24]. The errors used for validation are the root mean square error of the disparity (RMSE) and the mean absolute value of the relative error (REL), see Sec. 4.1. Note that DeiT3-L-22K-1K is DeiT3-L pretrained on ImageNet-22k and fine-tuned on ImageNet-1K, Next-ViT-L-1K is the shortened form of Next-ViT-L ImageNet-1K and Next-ViT-L-1K-6M stands for Next-ViT-L ImageNet-1K-6M. The model in italics is a retrained legacy model from MiDaS v3.0. The rows are ordered such that better models are at the top. The best numbers per column are bold and second best underlined.